

Question 1: What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:

Simple Linear Regression (SLR) is a statistical technique used to model the relationship between one independent variable (X) and one dependent variable (Y). It fits a straight line through the data points to predict the value of the dependent variable based on the independent variable.

Purpose:

- To understand the relationship between two variables.
 - To predict future values of the dependent variable.
 - To measure the impact of the independent variable on the dependent variable.
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Question 2: What are the key assumptions of Simple Linear Regression?

Answer:

The key assumptions are:

1. **Linearity** – The relationship between X and Y is linear.
 2. **Independence** – Observations are independent of each other.
 3. **Homoscedasticity** – The variance of residuals remains constant.
 4. **Normality** – Residuals are normally distributed.
 5. **No Significant Outliers** – Extreme values should not unduly influence the model.
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Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Where:

- **Y** = Dependent variable (target variable)
 - **X** = Independent variable (predictor variable)
 - **β_0** = Intercept (value of Y when X = 0)
 - **β_1** = Slope or regression coefficient
 - **ε** = Error term (residual)
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Question 4: Provide a real-world example where simple linear regression can be applied.

Answer:

A company may use simple linear regression to predict **sales revenue** based on **advertising expenditure**. Here:

- Advertising expenditure is the independent variable (X).
- Sales revenue is the dependent variable (Y).

As advertising spending increases, sales may also increase, and SLR can help estimate that relationship.

Question 5: What is the method of least squares in linear regression?

Answer:

The method of least squares is a technique used to find the best-fitting regression line by minimizing the sum of the squared differences between the actual values and the predicted values. These differences are called residuals.

The objective is to minimize:

$$\sum (Y_i - \hat{Y}_i)^2$$

This ensures the regression line fits the data as closely as possible.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:

Logistic Regression:

Logistic Regression is a supervised learning algorithm used for classification problems. It predicts the probability of an outcome belonging to a particular class.

Differences:

Linear Regression	Logistic Regression
Used for prediction of continuous values	Used for classification
Output can be any real number	Output ranges from 0 to 1
Uses a straight-line equation	Uses the sigmoid function
Predicts numerical values	Predicts class probabilities

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Answer:

1. Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

2. Mean Squared Error (MSE)

Measures the average of squared differences between actual and predicted values.

3. Root Mean Squared Error (RMSE)

The square root of MSE. It is expressed in the same unit as the target variable and penalizes larger errors more heavily.

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:

R-squared (R^2) measures the proportion of variance in the dependent variable that is explained by the independent variable(s).

- $R^2 = 0$ means the model explains none of the variability.
- $R^2 = 1$ means the model explains all the variability.

A higher R^2 value generally indicates a better fit of the model to the data.

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

Intercept (β_0)

The predicted value of the dependent variable when the independent variable is zero.

Slope (β_1)

Represents the expected change in the dependent variable for a one-unit increase in the independent variable.

Example:

If the regression equation is:

$$\text{Salary} = 20000 + 5000 \times \text{Experience}$$

- Intercept (20000): Expected salary when experience is 0 years.
- Slope (5000): Salary increases by ₹5,000 for each additional year of experience.